ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 11

AGRICULTURAL TECHNOLOGY

PAPER 1

MARKS: 100 **TIME: 2 hours**

Topics: materials and welding, tractors and implements, electricity and energy, water supply and structures.

Written by Academic Central for practice, following the CAPS curriculum for Grade 11. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions. Show formulae, substitutions and units in ALL calculations. Use g = 9,8 $m \cdot s^{-2}$ and the density of water = 1 000 $kg \cdot m^{-3}$.

QUESTION 1

1.1 Choose the correct answer. Write only the letter (A–D).

1.1.1 MIG welding protects the weld with a shielding gas such as …

A argon or carbon dioxide. B oxygen.
C acetylene. D hydrogen.

1.1.2 The power take-off (PTO) of a tractor is used to …

A drive implements such as a rotavator. B steer the tractor.
C brake the tractor. D inflate the tyres.

1.1.3 The SI unit of pressure is the …

A newton. B joule.
C pascal. D watt.

1.1.4 The main fuel gas in biogas is …

A hydrogen. B methane.
C butane. D oxygen.

1.1.5 An electric fence energiser sends …

A short, high-voltage pulses. B a constant low current.
C radio signals. D direct current at 12 V only.

(5 × 2)
(10)

1.2 Give ONE term for each description.

1.2.1 A system that uses oil under pressure to transmit force, e.g. to lift implements

1.2.2 The vertical height through which a pump must lift water

1.2.3 A plastic that softens when heated and can be reshaped many times

1.2.4 Servicing a tractor at fixed intervals (hours) before faults occur

1.2.5 A device that increases or decreases AC voltage (5 × 2)
(10)

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QUESTION 2: MATERIALS AND WELDING

2.1 Distinguish between thermoplastics and thermosetting plastics. Give ONE example of each. (4)

2.2 Give THREE uses of plastics on a farm. (3)

2.3 Give ONE advantage of MIG welding and ONE advantage of arc (stick) welding. (4)

2.4 Name TWO welding defects and give a cause of each. (4)

2.5 Why is thick cast iron preheated before it is welded? (3)

2.6 Name TWO ways to test a weld without destroying it. (2)

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QUESTION 3: TRACTORS AND IMPLEMENTS

3.1 Name FOUR daily checks before starting a tractor. (4)

3.2 State Pascal's principle and explain how a tractor's hydraulic system uses it to lift an implement. (4)

3.3 Give THREE safety rules when working with a PTO-driven implement. (3)

3.4 A tractor ploughs 6 ha per day and uses 12 ℓ of diesel per hectare. Diesel costs R23 per litre. For a 45 ha field, calculate: (a) the number of days needed; (b) the cost of the diesel. (5)

3.5 Why must tractor tyres be kept at the correct pressure? (2)

3.6 Name TWO implements used for primary tillage. (2)

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QUESTION 4: ELECTRICITY AND ENERGY

4.1 Name FOUR components of an electric fence system. (4)

4.2 Why is a good earth system important for an electric fence? (2)

4.3 A 230 V water heater in a dairy draws a current of 8,7 A. Calculate: (a) its power; (b) the energy used in 5 hours, in kWh; (c) the cost of this energy at R2,80 per kWh. (6)

4.4 Explain how a biogas digester works and name its TWO useful products. (5)

4.5 Give TWO advantages of using a solar-powered water pump. (2)

4.6 What is the purpose of an earth leakage unit? (1)

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QUESTION 5: WATER AND STRUCTURES

5.1 A centrifugal pump has a suction head of 4 m and a delivery head of 26 m. Calculate the total static head. (2)

5.2 Calculate the water pressure at the bottom of a pipe holding a column of water 30 m high. Use $p = \rho g h$ and give your answer in kPa. (3)

5.3 Give ONE advantage each of centre-pivot irrigation and drip irrigation. (4)

5.4 Why is the water–cement ratio important when making concrete? (3)

5.5 Why is steel mesh placed in a concrete water trough? (2)

5.6 Give THREE factors to consider when choosing a site for a farm dam. (3)

5.7 Why must the suction pipe of a pump be completely airtight? (3)

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TOTAL: 100

END OF QUESTION PAPER